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Analysis of the sPHENIX Photon and Jet Triggering System, Run 24 p+p Collisions at 200 GeV

Justin Bennett

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1 Introduction

The sPHENIX experiment is optimized for high-rate jet and photon measurements, relying on a fast data acquisition (DAQ) and trigger system to select rare high-p_T processes in heavy-ion collisions. At the heart of this system is a two-tier Level-1 trigger architecture [1]. First, multiple Local Level-1 (LL1) modules each generate a primitive yes/no decision for a specific physics object (e.g. a high-energy photon in the electromagnetic calorimeter (EMCal), or an energetic jet using both EMCal and hadronic calorimeter data for example). These LL1 decisions, along with timing and DAQ status signals, are passed every bunch crossing to a central Global Level-1 (GL1) trigger. Whenever the GL1 detects at least one satisfied LL1 trigger bit and the DAQ is not busy, it issues a Level-1 accept and initiates full detector readout. This mechanism ensures that the targeted high-p_T processes are recorded, while preventing triggers from firing when front-end electronics are busy or prescale settings require limiting the rate [2].

A crucial component of EMCal-based analyses is the photon trigger. The EMCal is divided into η-ϕ towers of size 0.025×0.025, each corrected for gain through lookup tables. These towers are clustered into non-overlapping 8×8 regions, and any region exceeding a preset threshold is flagged by the photon LL1. Similarly, jet triggers identify high-energy deposits in both EMCal and hadronic calorimeter data. In many runs, these photon or jet triggers are combined with a minimum-bias trigger from the minimum bias detector, $MBD\ NS \geq 1$, requiring at least one Photomultiplier Tube (PMT) signal to fire in coincidence in the north and south MBD.

Each active GL1 trigger is monitored by three counters:

- **Raw**: increments on every trigger occurrence,
- **Live**: increments only when the DAQ is ready to accept data (i.e. not busy),
- **Scaled**: increments according to a *scaledown factor* (i.e. the number of live events skipped).

For example, if the scaledown factor is 5, then the **Scaled** counter fires once for every 6 **Live** occurrences. If the scaledown is 0, the **Scaled** counter fires every time the **Live** counter does (i.e. no prescaling). A special exception is trigger 0, which is used for counting beam clocks rather than physics triggers; assigning it a zero scaledown would make it fire continuously at the beam clock rate, overwhelming the system.

In high-luminosity environments, scaledown factors play a central role in balancing the acceptance of frequently firing triggers against DAQ bandwidth limits. Quantitatively,

$$\text{Scaledown Factor} + 1 \;=\; \frac{\text{live counts}}{\text{scaled counts}},$$

indicating that the number of fully recorded events is reduced by the scaledown value. Because scaledown settings can vary across runs, applying run-by-run scaledown corrections is essential for accurately interpreting trigger rates and efficiencies. This is especially important for producing reliability checks and efficiency plots, ensuring that measured data properly reflect the true rate for critical photon and jet events.

In summary, the synergy between the LL1 and GL1 trigger layers, along with the raw/live/scaled counters and flexible scaledown factors, provides sPHENIX with a robust and efficient framework

for capturing key high-energy processes. This design enables the experiment to meet demanding rate requirements, collect high-p_T jets and photons, and maintain precise control over trigger bandwidth in a high-luminosity environment.

2 Efficiency Analysis and Results

2.1 Data Selection

This analysis defines a consistent dataset by applying several selection criteria to each run:

- **Event Count Requirement:** Runs must contain at least 1M events and have a run number ≥ 47289.
- **Calorimeter QA:** The EMCal, IHCal, and OHCal must each be flagged as "Golden" by the QA process.
- **Minimum Runtime:** Runs must be longer than five minutes in duration.
- **MBD Livetime:** The MBD (minimum-bias) livetime must exceed 80%.
- **Magnet-On Requirement (Default):** Only runs taken with the magnet powered on are included.

In addition, the data are grouped according to whether each trigger was active or inactive, determined by its scaledown factor in the GL1 database. Specifically, if a scaledown factor is set to -1, that trigger is considered inactive for the given run. Table 1 summarizes the triggers analyzed here, showing the number of runs in which each trigger is active (*ON*) or inactive (*OFF*), along with the corresponding average prescale value.

For photon triggers at thresholds of 3, 4, and 5 GeV, the scaled bits were jointly active in 643 runs, enabling a shared efficiency analysis over those runs. By contrast, when analyzing only the 4 and 5 GeV photon triggers, 648 runs are available, providing a slightly larger dataset. This differs from the jet triggers, which are all active for the same 646 runs. In each case, the minimum-bias condition, MBD N&S ≥ 1, is also included and may be subject to larger prescales due to its high firing rate.

2.2 Photon Triggers

The distribution presented in Fig. 1 is obtained by combining the MBD NS ≥ 1 (MBD NS) raw bits with the live bits of the Photon $3, 4, 5$ GeV + MBD NS ≥ 1 triggers. Specifically, each event is first required to fire the MBD NS raw bit, after which the maximum cluster energy is recorded in a histogram associated with that trigger. In the same event, the live bits of each photon trigger are checked; if any are active, the corresponding photon-trigger histograms are also filled. This method circumvents the need for run-by-run prescale corrections that would otherwise be necessary when using only scaled trigger bits.

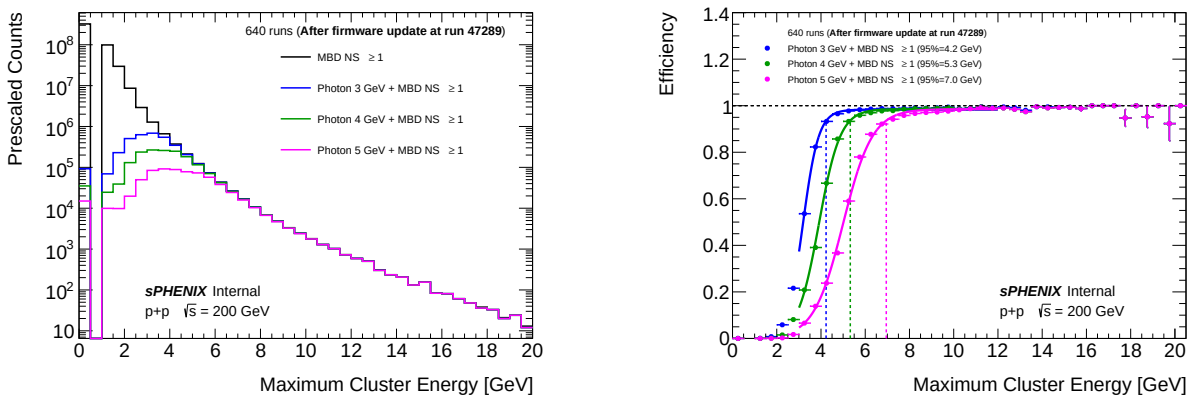
Trigger Name	Runs ON	Runs OFF	Ave Prescale
MBD N&S ≥ 1	868	0	2004.702
Photon 3 GeV + MBD NS ≥ 1	643	225	4.754
Photon 4 GeV + MBD NS ≥ 1	648	220	1.084
Photon 5 GeV + MBD NS ≥ 1	648	220	1.067
Jet 8 GeV + MBD NS ≥ 1	646	222	1.024
Jet 10 GeV + MBD NS ≥ 1	646	222	1.004
Jet 12 GeV + MBD NS ≥ 1	646	222	1.001

Table 1: Combined summary of photon and jet triggers, run counts, and average prescale values.

The resulting efficiency distributions are fit using the automated procedure described in Sec. 3, which for this nominal fit, employs the logistic function

$$f(x) = \frac{p_0}{1 + e^{(-p_1(x-p_2))}} \tag{1}$$

where p_0 is the amplitude (asymptotic plateau), p_1 is the slope, and p_2 is the horizontal offset. As indicated in the legend of the efficiency curve in Fig. 1, the photon 3, 4, and 5 GeV + MBD NS triggers reach 95% efficiency relative to the MBD NS trigger at 4.2, 5.4, and 7.0 GeV, respectively.



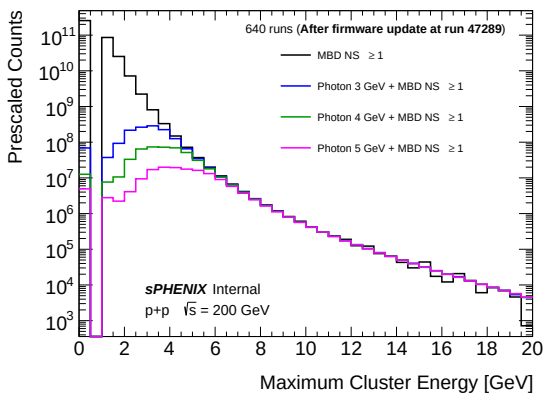
(a) Maximum Cluster Energy

(b) Photon 3,4,5 Trigger/MBD NS

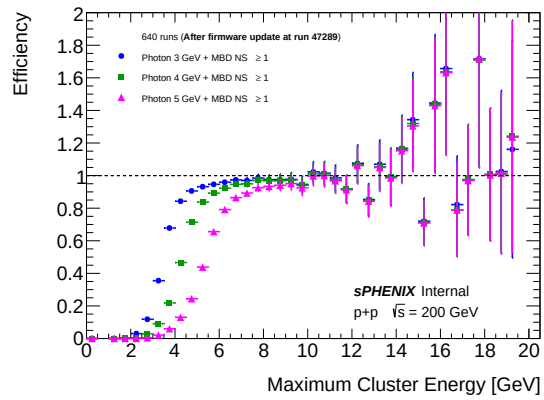
Figure 1: Raw MBD NS and live rare trigger logic photon efficiency curves for Photon 3, 4, 5 GeV triggers. Using Raw MBD NS Bits and live bits for the rare triggers. Only if the raw MBD NS bit fired, is a rare trigger's histogram filled. The error bars correspond to a binomial error calculation automated by ROOT.

In Fig. 2, the distributions are obtained using only the active GL1 scaled bits. Run-by-run scaledown factors are then applied offline, based on total live and scaled counts retrieved from the GL1 database. Although the two trigger-logic methods used to fill the maximum cluster energy histograms differ, taking their ratio and plotting the resulting efficiency distribution provides a cross-check that the efficiency curves are qualitatively consistent when calculated in different ways. Specifically, in subfigures (a) and (b) of Fig. 2, the efficiency is determined using the maximum cluster energy in each event, whereas in subfigures (c) and (d) it is determined using the maximum

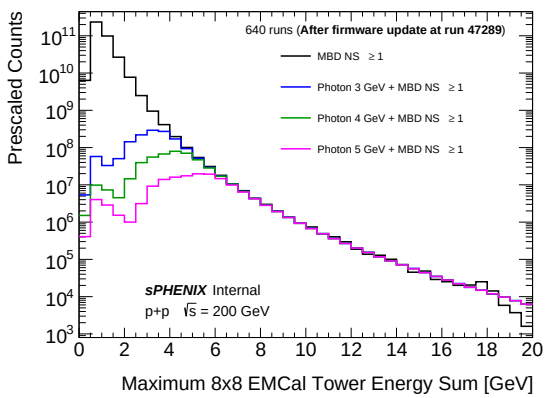
energy in a single interface board (an 8×8 tower sum in the EMCal). In both cases, the resulting efficiency curves agree with one another and match those shown in Fig. 1.



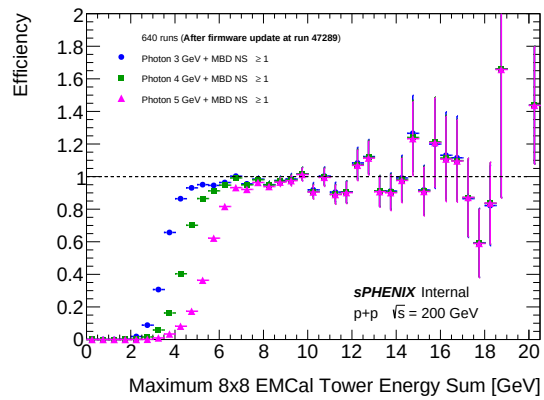
(a) Maximum Cluster Energy - Scaled Bits Only



(b) Photon Trigger Efficiency vs. MBD (NS)



(c) Max 8×8 Board Energy - Scaled Bits Only

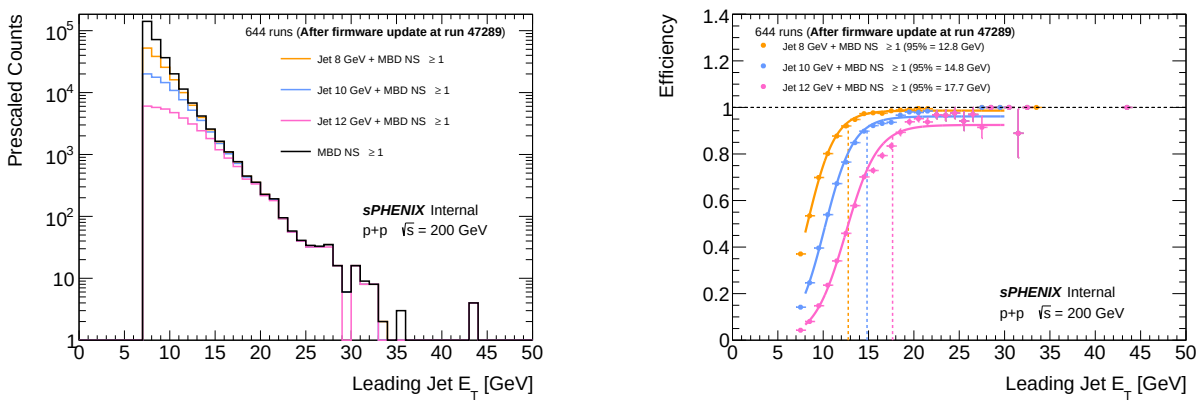


(d) Photon Trigger Efficiency vs. MBD (NS)

Figure 2: Combined photon efficiency curves for the 3, 4, and 5 GeV photon triggers. **Top row:** maximum cluster energy (left) and the corresponding trigger turn-on (right). **Bottom row:** maximum energy in an EMCal 8×8 board (left) and its trigger turn-on (right). All panels use the same scaled MBD (NS) and rare-triggers reference.

2.3 Jet Triggers

The jet efficiency curves in Fig. 3 are fit with the same logistic function from Eq. (1) used for the photon trigger. Under these conditions, the Jet 8, 10, and 12 GeV + MBD NS ≥ 1 triggers reach 95% efficiency relative to the MBD NS trigger at 12.8, 14.8, and 17.7 GeV, respectively. Note that the logic for filling the jet-trigger histograms matches that of Fig. 1, employing raw MBD NS bits and live rare jet trigger bits rather than relying on scaled trigger bits with offline run-by-run scaling factors.



(a) Leading Jet E_T

(b) Jet X Trigger/MBD NS

Figure 3: Jet efficiency curves for jet 8, 10, 12 GeV triggers.

Additionally, Fig. 4 shows a summary of the turn-on points at which each trigger's efficiency curve, as determined by the nominal logistic fit in Eq. 1, reaches 95% of its fitted amplitude.

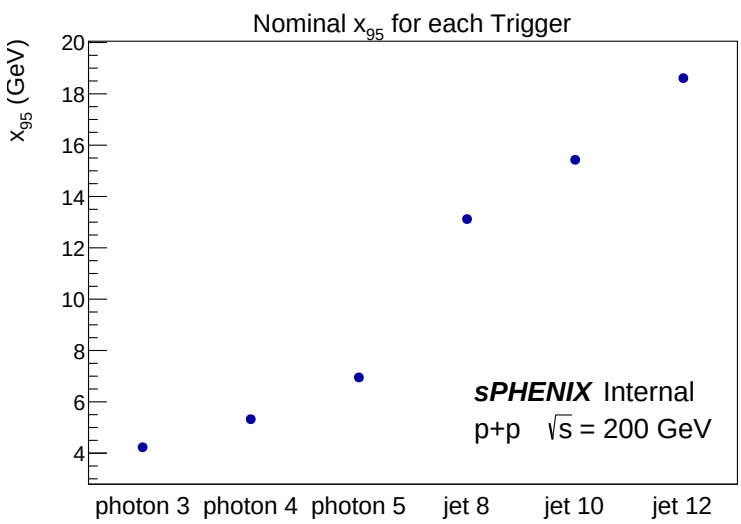


Figure 4: Distribution of the 95% efficiency thresholds for each trigger using the nominal logistic fit.

Relevant TF1 fit functions and histograms for jet and photon triggers are available in the GitHub repository at https://github.com/JustinBen0131/DirectPhotonAnalysis. The files of interest are:

- `FitFunctions_Jet_X_GeV_plus_MBD_NS_geq_1.root` (where $X = 8, 10, 12$)
- `FitFunctions_Photon_X_GeV_plus_MBD_NS_geq_1.root` (where $X = 3, 4, 5$)

Within each file, a folder is named after the corresponding trigger. Inside this folder, subdirectories correspond to each fit function type outlined in Section 3. Each subdirectory contains a `TF1` representing the final fitted efficiency curve and the associated histogram for that trigger.

3 Fit Analysis

To quantify the turn-on behavior of photon triggers relative to a minimum-bias reference, several functional forms are considered. This section introduces the mathematical representations of these models, describes how the efficiency threshold is determined, and outlines the procedure for parameter initialization and iterative fitting. Each model is explored in both three- and four-parameter versions, using either *log-slope* or *simple-slope* parameterizations. The primary objective is to achieve robust, stable fits across a wide range of efficiency histograms, and to accommodate variations that inform systematic uncertainties.

Fit Strategy and Varied Models

Three principal functional forms are explored: **Logistic**, **Error Function (erf)**, and **Gumbel**. Each can be realized in a three- or four-parameter configuration, depending on whether a separate baseline or asymptote is explicitly included. In addition, two conventions are employed for the slope parameter:

Log-Slope (default) Uses $\alpha = \ln(\text{slope})$, ensuring positive slopes and improving numerical stability across a large dynamic range.

Simple-Slope (alternative) Fits the slope directly as a linear coefficient, providing a more transparent parameter but sometimes resulting in unstable fits if the slope varies by orders of magnitude.

The fitting procedure begins with an automated initialization phase that determines approximate amplitude, slope, and offset values from the efficiency histogram (targeting reference points near 20%, 50%, and 80% of its final plateau). In the *log-slope* approach, $\ln(\text{slope})$ is fitted directly to ensure a positive slope and improve numerical stability; alternatively, a *simple-slope* method treats the slope as a linear coefficient.

Following this initialization, the fit is carried out using a minimization algorithm (e.g. Minuit2) and can iterate up to 20 times, terminating early if the estimated distance to the minimum (EDM) falls below 10^{-9}. If the algorithm reaches the iteration limit without meeting the EDM criterion, it retains the best solution found. In some cases, the fit is repeated (re-initialized at the best-fit parameters) for additional refinement. This multi-stage approach ensures stable and precise parameter estimation even when combining data from hundreds of runs, some with minor inconsistencies that can accumulate in the final distributions. By iteratively refining the fit, the procedure derives an appropriate average across these variations, which is vital for determining the 95% efficiency threshold relative to the MBD NS trigger.

Table 2 associates each letter label (A–I) with a specific parametric form used in Tables 4 5. Each model can adopt either a *log-slope* ($\alpha = \ln(\text{slope})$) or *simple-slope* (slope) parameterization. Three-parameter versions feature a single amplitude (or plateau) and an offset, while four-parameter variants add a second asymptote or baseline.

Table 2: Overview of function keys, their textual identifiers, and the corresponding mathematical definitions. Each model may use $\alpha = \ln(\text{slope})$ (*log-slope*) or a direct slope (*simple-slope*).

Key	Function Name	Mathematical Form
A	`logisticFreeAmp`	$f_A(x) = \dfrac{\text{Amp}}{1 + e^{-e^{\alpha}\left(x - x_{\text{off}}\right)}}$
B	`erfFreeAmp`	$f_B(x) = 0.5\,\text{Amp}\left[1 + \text{erf}\left(\dfrac{e^{\alpha}\left(x - x_{\text{off}}\right)}{\sqrt{2}}\right)\right]$
C	`gumbelFreeAmp`	$f_C(x) = \text{Amp}\, e^{-e^{-e^{\alpha}\left(x - x_{\text{off}}\right)}}$
D	`logistic4Param`	$f_D(x) = \ell + \dfrac{h - \ell}{1 + e^{-e^{\alpha}\left(x - x_{\text{off}}\right)}} \quad (\ell \equiv p_0,\ h \equiv p_1)$
E	`erf4Param`	$f_E(x) = \text{base} + \text{scale}\,\dfrac{1}{2}\left[1 + \text{erf}\left(\dfrac{e^{\alpha}\left(x - x_{\text{off}}\right)}{\sqrt{2}}\right)\right]$
F	`gumbel4Param`	$f_F(x) = \text{base} + \text{scale}\, e^{-e^{-e^{\alpha}\left(x - x_{\text{off}}\right)}}$
G	`logisticSimpleAmp`	$f_G(x) = \dfrac{\text{Amp}}{1 + e^{-(\text{slope})\left(x - x_{\text{off}}\right)}}$
H	`erfSimpleAmp`	$f_H(x) = 0.5\,\text{Amp}\left[1 + \text{erf}\left(\dfrac{(\text{slope})\left(x - x_{\text{off}}\right)}{\sqrt{2}}\right)\right]$
I	`gumbelSimpleAmp`	$f_I(x) = \text{Amp}\, e^{-e^{-(\text{slope})\left(x - x_{\text{off}}\right)}}$

Table 3: Analytic expressions for the 95% crossing x_{95} when the model output reaches 0.95 of its effective plateau. Each expression uses the same parameter definitions as in Table 2.

Function Name	Expression for x_{95}
`logisticFreeAmp`	$x_{95} = x_{\text{off}} + \frac{1}{e^{\alpha}} \ln\left(\frac{0.95}{1-0.95}\right)$
`erfFreeAmp`	$x_{95} = x_{\text{off}} + \frac{\sqrt{2}}{e^{\alpha}} \operatorname{erf}^{-1}(0.90)$
`gumbelFreeAmp`	$x_{95} = x_{\text{off}} + \frac{1}{e^{\alpha}} \left\{ -\ln\left[-\ln\left(\frac{0.95}{\text{Amp}}\right)\right]\right\}$
`logistic4Param`	$f_{\text{D}}(x_{95}) = \ell + 0.95\,(h-\ell) \implies x_{95}$ solves $f_{\text{D}}(x_{95}) = \dots$
`erf4Param`	$f_{\text{E}}(x_{95}) = \text{base} + 0.95\,(\text{scale}) \times \frac{1}{2} \implies x_{95}$ solves $f_{\text{E}}(x_{95}) = \dots$
`gumbel4Param`	$f_{\text{F}}(x_{95}) = \text{base} + 0.95\,(\text{scale}) \implies x_{95}$ solves $f_{\text{F}}(x_{95}) = \dots$
`logisticSimpleAmp`	$x_{95} = x_{\text{off}} - \frac{1}{\text{slope}} \ln\left(\frac{0.95}{1-0.95}\right)$
`erfSimpleAmp`	$x_{95} = x_{\text{off}} + \frac{\sqrt{2}}{\text{slope}} \operatorname{erf}^{-1}(0.90)$
`gumbelSimpleAmp`	$x_{95} = x_{\text{off}} + \frac{1}{\text{slope}} \left\{ -\ln\left[-\ln\left(\frac{0.95}{\text{Amp}}\right)\right]\right\}$

3.1 Photon Triggers

Below are 3×3 tables for each photon triggers efficiency distribution, where each distribution corresponds to one of the previously defined fit functions. Table 4 presents the quantitative fit results for these distributions, and the corresponding fits for each variation are illustrated in Figs. 5, 6, and 7.

Table 4: Condensed fit results for three photon triggers (3 GeV, 4 GeV, 5 GeV), each combined with `MBD_NS_geq_1`. Each row lists the method label, χ^2/NDF, x_{95} (GeV), and model parameters. The Slope (or $\ln s$) column shows the slope for 3-parameter fits or $\alpha = \ln(\mathrm{slope})$ for 4-parameter fits. All numeric values have three significant figures in the central value and two in the uncertainty.

Fit Fun	χ^2/NDF	x_{95}	**Amplitude**	**Slope (or $\ln s$)**	**Low**	**High**	**Base**	**Scale**	**Offset**
				Photon 3 GeV + MBD NS $\geq$ 1					
A	58.3	4.23	0.98± 0.0002	1.0± 0.0003	NA	NA	NA	NA	3.2± 0.0007
B	177	4.17	0.98± 0.0002	0.50± 0.0002	NA	NA	NA	NA	3.2± 0.0008
C	22.4	4.28	0.99± 0.0002	0.88± 0.0003	NA	NA	NA	NA	3.0± 0.0009
D	14.9	3.43	NA	0.76± 0.002	−5.4± 0.07	0.99± 0.0002	NA	NA	2.1± 0.006
E	24.6	1.66	NA	−0.44± 0.001	NA	NA	−120± 0.002	120± 0.002	−0.89± 0.005
F	14.9	3.75	NA	0.76± 0.003	NA	NA	−2.2± 0.002	3.2± 0.002	2.4± 0.002
G	58.3	4.23	0.98± 0.0002	2.8± 0.007	NA	NA	NA	NA	3.2± 0.0007
H	177	4.17	0.98± 0.0002	1.7± 0.004	NA	NA	NA	NA	3.2± 0.0008
I	22.4	4.42	0.99± 0.0002	2.4± 0.006	NA	NA	NA	NA	3.0± 0.0009
				Photon 4 GeV + MBD NS $\geq$ 1					
A	183	5.32	0.99± 0.0004	0.73± 0.002	NA	NA	NA	NA	3.9± 0.0009
B	225	5.21	0.98± 0.0003	0.23± 0.002	NA	NA	NA	NA	3.9± 0.0009
C	937	5.73	1.0± 0.0004	0.33± 0.002	NA	NA	NA	NA	3.6± 0.0007
D	47.1	5.20	NA	0.91± 0.004	0.092± 0.002	0.98± 0.0004	NA	NA	4.0± 0.002
E	120	5.11	NA	0.41± 0.004	NA	NA	0.10± 0.002	0.87± 0.002	4.0± 0.002
F	23.8	5.37	NA	0.71± 0.003	NA	NA	0.19± 0.0006	0.79± 0.0008	3.9± 0.001
G	183	5.32	0.99± 0.0004	2.1± 0.003	NA	NA	NA	NA	3.9± 0.0009
H	225	5.21	0.98± 0.0003	1.3± 0.002	NA	NA	NA	NA	3.9± 0.0009
I	937	5.71	1.0± 0.0004	1.4± 0.002	NA	NA	NA	NA	3.6± 0.0007
				Photon 5 GeV + MBD NS $\geq$ 1					
A	155	6.95	0.99± 0.0006	0.41± 2	NA	NA	NA	NA	5.0± 0.0007
B	259	6.89	0.98± 0.0008	−0.14± 0.002	NA	NA	NA	NA	5.0± 0.002
C	1170	8.38	1.0± 0.0009	−0.24± 0.002	NA	NA	NA	NA	4.6± 0.002
D	154	6.88	NA	0.45± 0.004	0.010± 0.001	0.98± 0.0008	NA	NA	5.0± 0.002
E	172	6.70	NA	−0.019± 0.003	NA	NA	0.029± 0.0006	0.95± 0.001	5.0± 0.002
F	493	7.58	NA	0.035± 0.003	NA	NA	0.059± 0.0004	0.95± 0.001	4.7± 0.002
G	155	6.95	0.99± 0.0008	1.5± 0.003	NA	NA	NA	NA	5.0± 0.002
H	259	6.89	0.98± 0.0008	0.87± 0.001	NA	NA	NA	NA	5.0± 0.002
I	1170	7.60	1.0± 0.0009	0.79± 0.001	NA	NA	NA	NA	4.6± 0.002

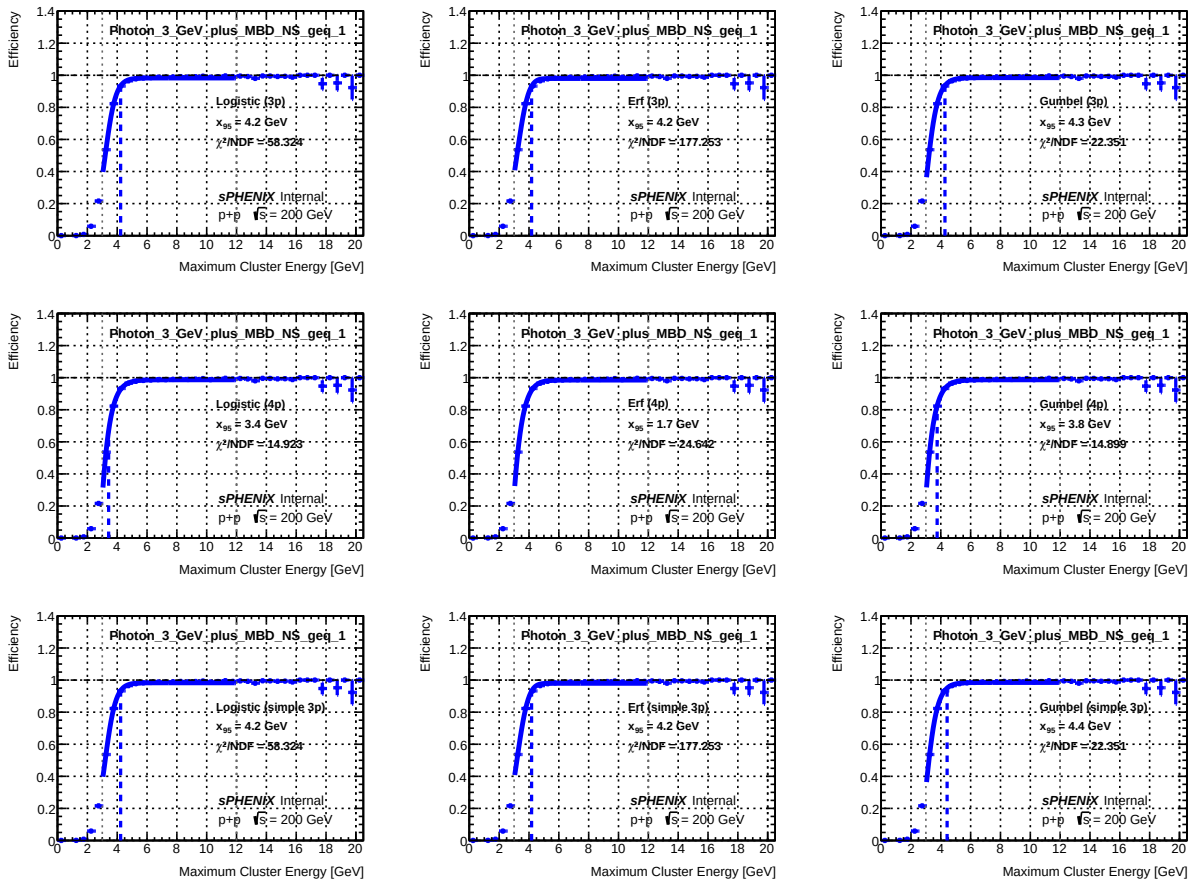


Figure 5: Fit variations for Photon 3 GeV + MBD NS $\geq$ 1 Efficiency Distributions.

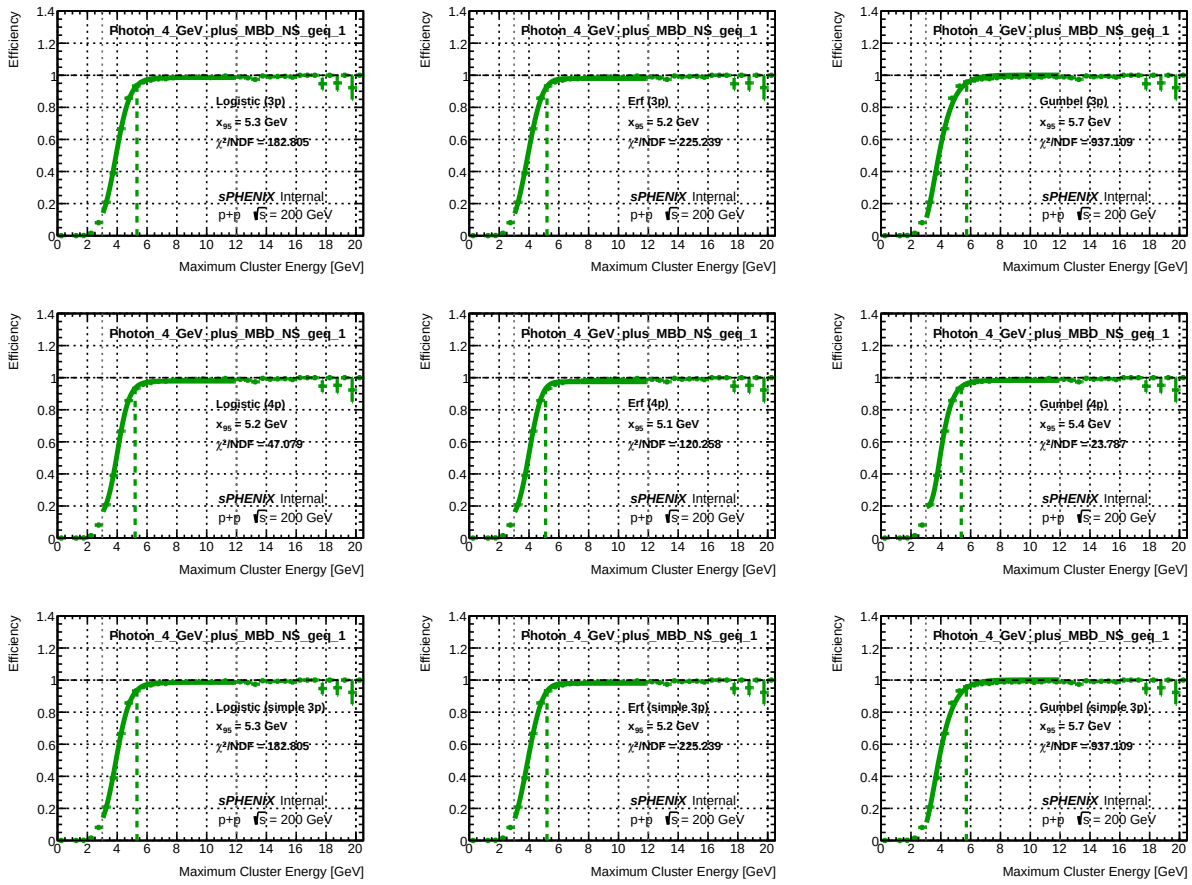


Figure 6: Fit variations for Photon 4 GeV + MBD NS $\geq$ 1 Efficiency Distributions.

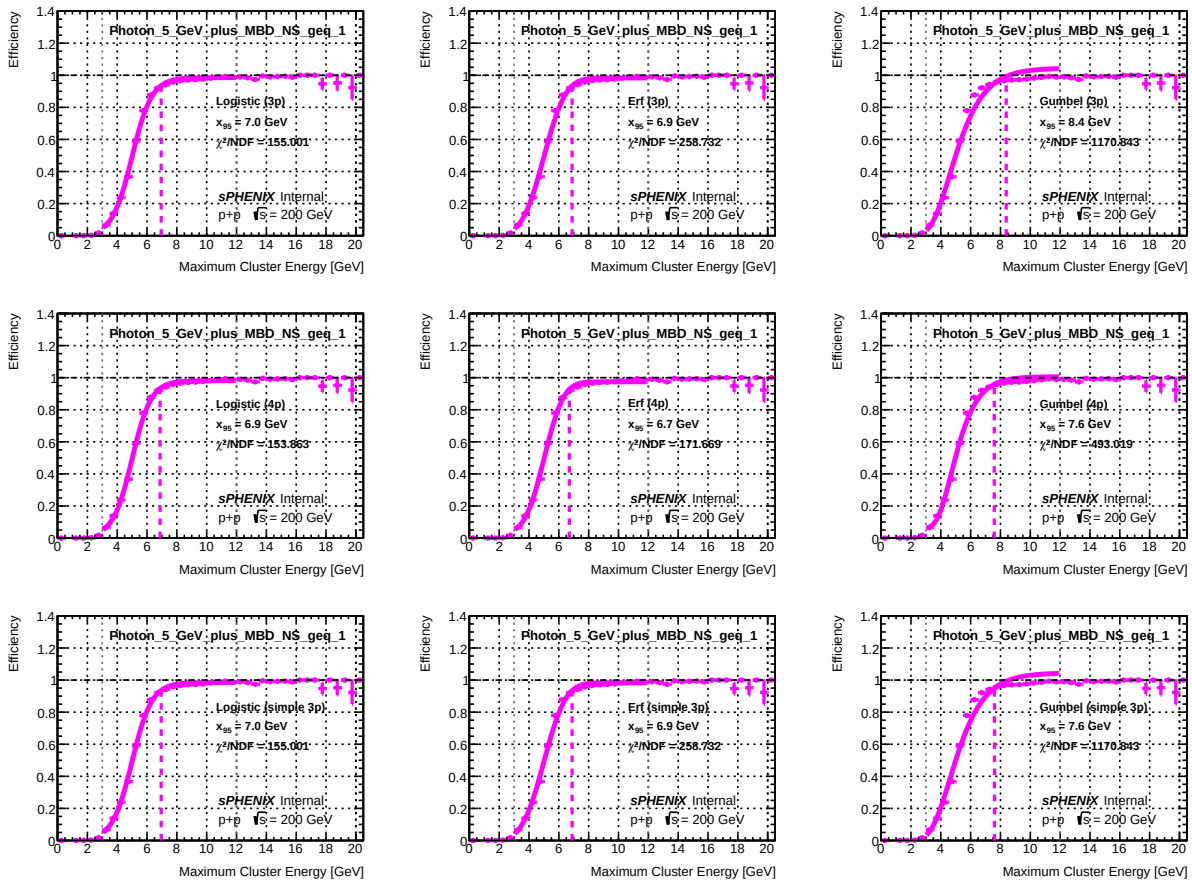


Figure 7: Fit variations for Photon 5 GeV + MBD NS $\geq$ 1 Efficiency Distributions.

3.2 Jet Triggers

Now can similarly organize the fit results for the jet triggers in table 5 below, with the corresponding fit for each variation on the following three pages in figures 8, 9, and 10.

Table 5: Condensed fit results for three jet triggers (8 GeV, 10 GeV, 12 GeV), each combined with MBD NS ≥ 1. Each row lists the method label, χ^2/NDF, x_{95} (GeV), and model parameters. The Slope (or $\ln s$) column shows the slope for 3-parameter fits or $\alpha = \ln(\mathrm{slope})$ for 4-parameter fits. All numeric values have three significant figures in the central value and two in the uncertainty.

Fit Fun	χ^2/NDF	x_{95}	**A**	**Slope (or $\ln s$)**	**Low**	**High**	**Base**	**Scale**	**Offset**
				Jet 8 GeV + MBD NS ≥ 1					
A	1.09	13.1	0.99± 0.002	−0.56± 0.04	NA	NA	NA	NA	8.0± 0.1
B	1.38	13.0	0.99± 0.002	−1.2± 0.04	NA	NA	NA	NA	7.5± 0.1
C	1.07	13.2	0.99± 0.002	−0.63± 0.04	NA	NA	NA	NA	7.6± 0.1
D	1.15	11.4	NA	−0.64± 0.03	−1.4± 0.3	0.99± 0.002	NA	NA	5.9± 0.3
E	1.17	2.06	NA	−1.9± 0.01	NA	NA	−110± 0.01	110± 0.01	−8.9± 0.3
F	1.15	13.2	NA	−0.62± 0.04	NA	NA	0.0025± 0.01	0.99± 0.01	7.6± 0.1
G	1.09	13.1	0.99± 0.002	0.57± 0.02	NA	NA	NA	NA	8.0± 0.1
H	1.38	13.0	0.99± 0.002	0.30± 0.01	NA	NA	NA	NA	7.5± 0.1
I	1.07	13.5	0.99± 0.002	0.54± 0.02	NA	NA	NA	NA	7.6± 0.1
				Jet 10 GeV + MBD NS ≥ 1					
A	1.43	15.4	0.97± 0.004	−0.60± 0.03	NA	NA	NA	NA	10± 0.03
B	2.33	15.0	0.97± 0.004	−1.1± 0.02	NA	NA	NA	NA	10± 0.03
C	1.19	15.9	0.98± 0.004	−0.78± 0.03	NA	NA	NA	NA	9.4± 0.04
D	1.29	14.7	NA	−0.76± 0.1	−0.62± 0.7	0.98± 0.006	NA	NA	8.4± 1
E	1.37	9.75	NA	−1.8± 0.02	NA	NA	−10± 0.02	11± 0.02	−0.32± 0.2
F	1.29	15.9	NA	−0.78± 0.09	NA	NA	0.017± 0.3	0.96± 0.3	9.4± 1
G	1.43	15.4	0.97± 0.004	0.55± 0.01	NA	NA	NA	NA	10± 0.03
H	2.33	15.0	0.97± 0.004	0.33± 0.008	NA	NA	NA	NA	10± 0.03
I	1.19	17.0	0.98± 0.004	0.46± 0.01	NA	NA	NA	NA	9.4± 0.04
				Jet 12 GeV + MBD NS ≥ 1					
A	2.99	18.6	0.95± 0.007	−0.70± 0.02	NA	NA	NA	NA	13± 0.05
B	3.49	18.0	0.94± 0.007	−1.2± 0.02	NA	NA	NA	NA	13± 0.05
C	2.19	20.8	0.99± 0.009	−1.1± 0.02	NA	NA	NA	NA	12± 0.04
D	2.28	19.3	NA	−0.91± 0.07	−0.18± 0.07	0.97± 0.01	NA	NA	12± 0.3
E	2.54	18.5	NA	−1.5± 0.08	NA	NA	−0.28± 0.1	1.2± 0.1	11± 0.5
F	2.05	20.3	NA	−1.0± 0.05	NA	NA	0.086± 0.03	0.89± 0.04	12± 0.2
G	2.99	18.6	0.95± 0.007	0.50± 0.01	NA	NA	NA	NA	13± 0.05
H	3.49	18.0	0.94± 0.007	0.31± 0.006	NA	NA	NA	NA	13± 0.05
I	2.19	21.4	0.99± 0.009	0.33± 0.007	NA	NA	NA	NA	12± 0.04

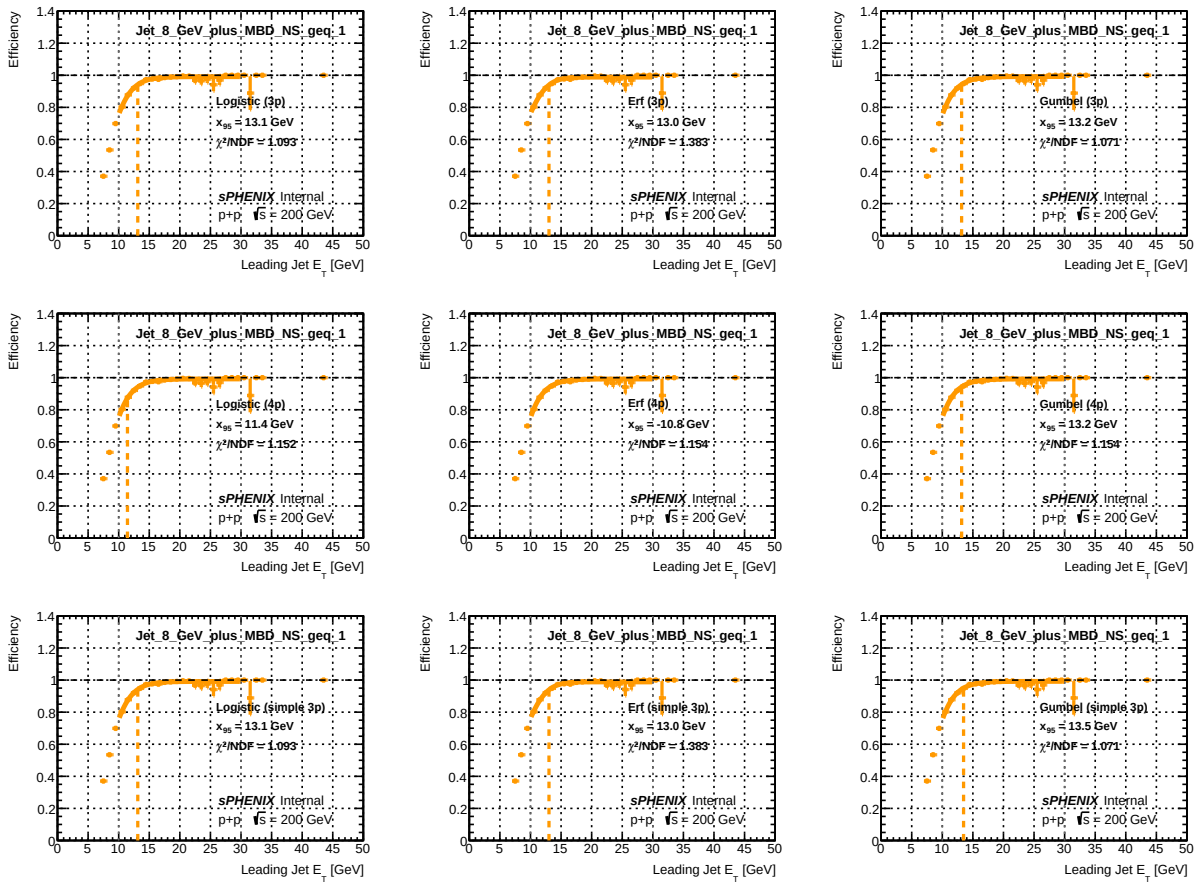


Figure 8: Fit variations for Jet 8 GeV + MBD NS $\geq$ 1 Efficiency Distributions.

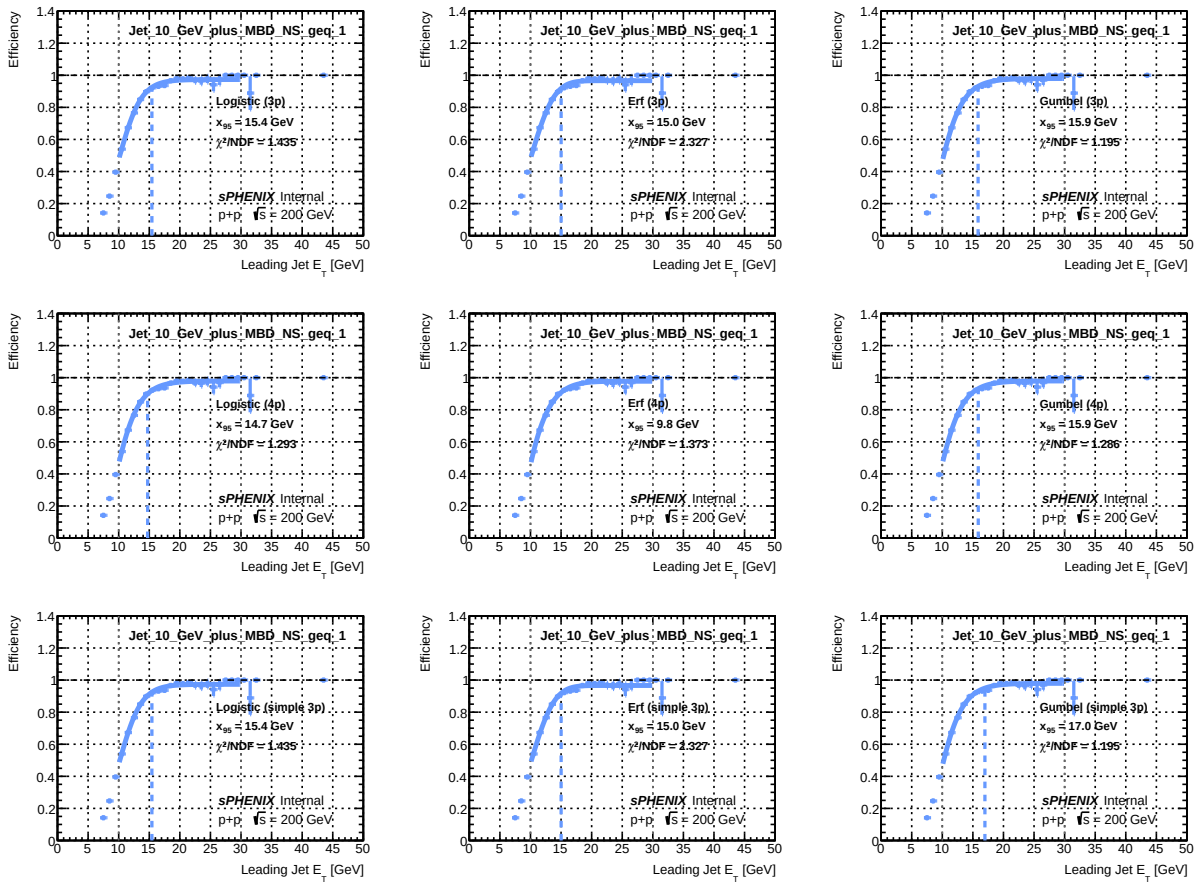


Figure 9: Fit variations for Jet 10 GeV + MBD NS $\geq$ 1 Efficiency Distributions.

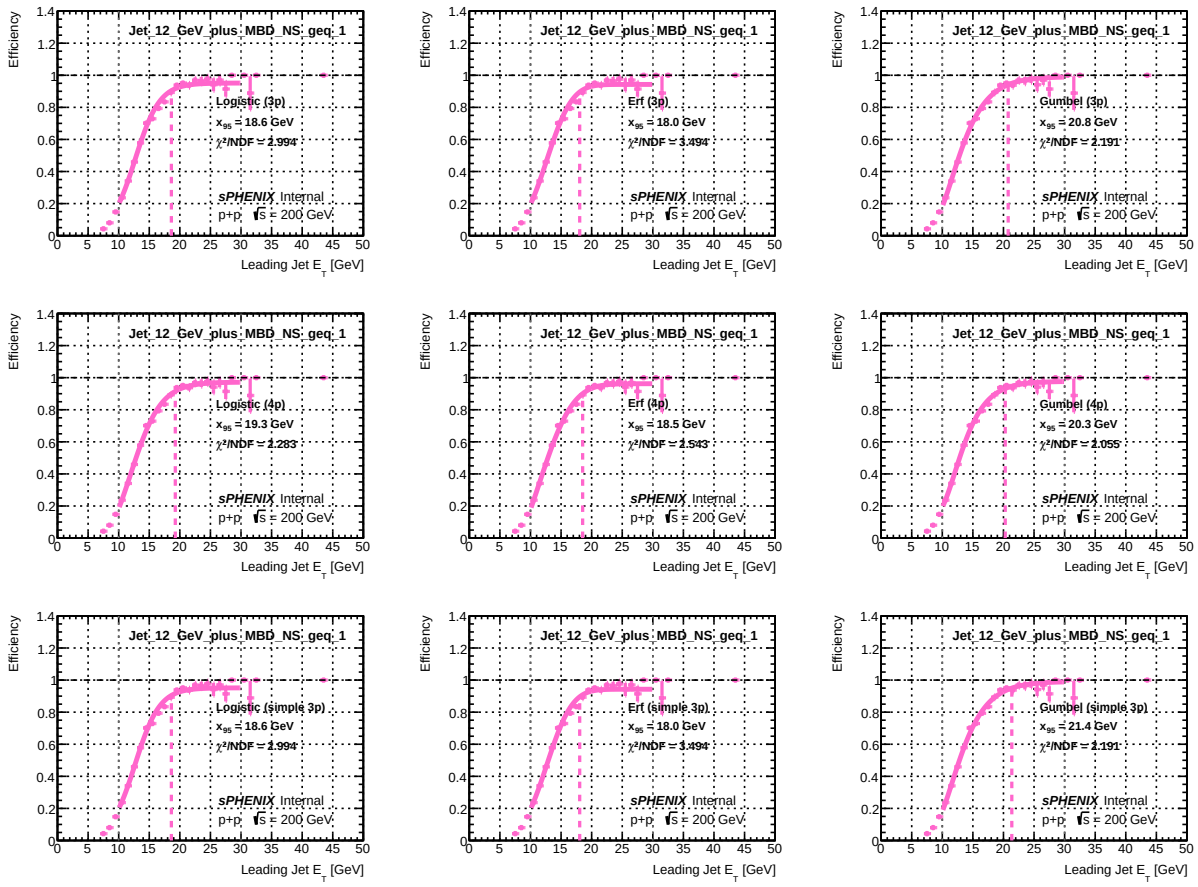


Figure 10: Fit variations for Jet 12 GeV + MBD NS $\geq$ 1 Efficiency Distributions.

4 Summary and Relative Efficiency Differences

A cross-check of the fitted 95% efficiency points for each parametric function was performed by examining χ^2/NDF (see Tables 4 and 5). Although smaller χ^2/NDF values typically indicate a closer match to the data, comparing different functional families is not entirely straightforward, because four-parameter fits (e.g. `logistic4Param`, `erf4Param`, `gumbel4Param`) can reshape tails more flexibly and thus introduce additional correlations.

Despite these nuances, certain models, such as `gumbelFreeAmp` and `logistic4Param`, consistently yield low χ^2/NDF for many photon triggers, implying a well described turn-on region. Others (`erfFreeAmp`, for instance) occasionally return higher χ^2 when confronted with asymmetric features not well captured by a Gaussian cumulative. At the same time, Figure 11 demonstrates that the 95% thresholds from all fits vary by only a few percent, underscoring that the measured turn-on region remains tightly constrained.

Nevertheless, certain edge cases do arise. For example, the `gumbelFreeAmp` fits at 5 GeV-shown in Fig. 7-exhibit visually poor agreement with the data in the high-energy tail, which inflates χ^2/NDF. This mismatch suggests that the Gumbel function may be less adaptable than other forms in modeling subtle features at higher energies, even though it still locates a 95% crossing within the same general range.

For photon triggers, `gumbelFreeAmp` often achieves comparatively low χ^2/NDF and places the 95% point close to that of `logistic4Param`. At 3 GeV, the χ^2/NDF for `gumbelFreeAmp` is 22.4 (compared to 14.9 for `logistic4Param`), yet the 95% crossing remains between 4.28 and 4.42 GeV. At 4 GeV, some four-parameter fits (e.g. 23.8 for `gumbel4Param`) return lower χ^2/NDF than simpler alternatives, but the resulting thresholds still fall within 5.2–5.7 GeV. At 5 GeV, `gumbelFreeAmp` and `logistic4Param` yield χ^2/NDF values of roughly 493 and 154, respectively, but both locate a 95% turn-on between 6.9 and 7.6 GeV.

Jet triggers display a similar pattern. For the 8 GeV trigger, `gumbelFreeAmp` ($\chi^2/\mathrm{NDF} = 1.07$) and `logisticSimpleAmp` ($\chi^2/\mathrm{NDF} = 1.09$) both point to a 95% efficiency near 13.1–13.5 GeV, while the four-parameter fits converge to thresholds in the same range. At higher jet thresholds (10 and 12 GeV), χ^2/NDF values vary among the fits, yet the extracted 95% crossings differ by only a few percent.

In summary, three-parameter functions such as `logisticSimpleAmp` and `gumbelFreeAmp` generally capture the core turn-on behavior while offering a simpler, more stable fit landscape than four-parameter models. The data typically rise from zero to a single plateau near unity, making any additional baseline parameter in the four-parameter formulations superfluous in many cases. By reducing the risk of overfitting and avoiding potential local minima, these simpler versions reliably constrain the 95% efficiency threshold within a few percent of their more flexible counterparts. Consequently, analyses centered on threshold extraction or systematic uncertainties can comfortably rely on simpler functions. However, in scenarios that demand explicit modeling of baseline offsets or asymmetries, four-parameter models may still provide key advantages—despite potential overfitting risks—by affording greater flexibility to match the datas subtler features.

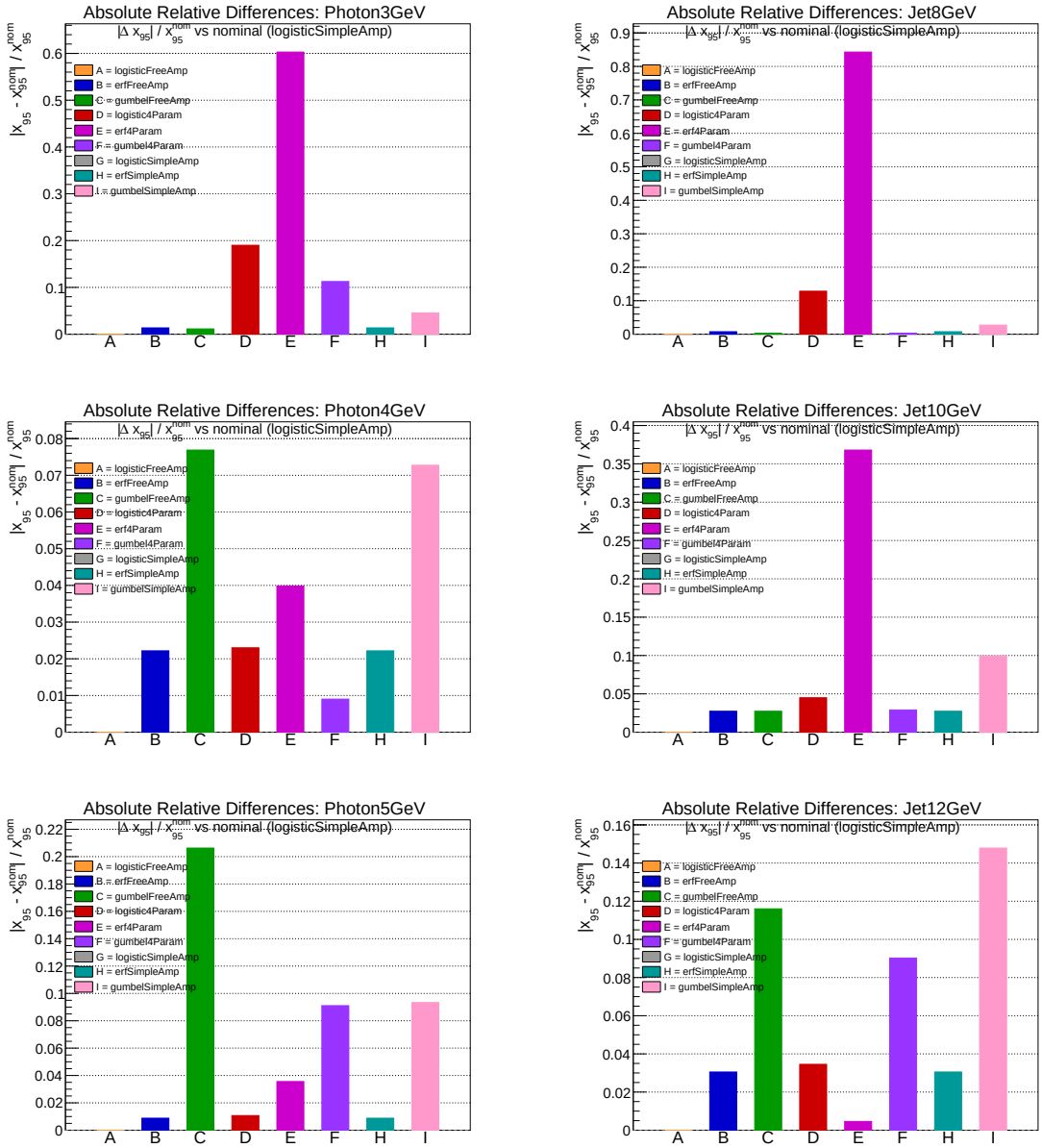


Figure 11: Absolute relative difference bar charts comparing `logisticSimpleAmp` (the nominal reference) to other fits. Photon triggers at thresholds 3, 4, and 5 GeV appear in the left column (top to bottom), and jet triggers at 8, 10, and 12 GeV are shown in the right column. The height of each bar quantifies $\left|x_{95}^{(\text{other})} - x_{95}^{(\text{nominal})}\right| / x_{95}^{(\text{nominal})}$. Even when certain four-parameter fits yield a lower χ^2/NDF, their 95% efficiency crossing remains within a few percent of the simpler three-parameter functions.

5 Future Work

While the analysis presented provides a reliable approximation of the efficiency points for both the photon and jet triggers, there remain aspects requiring further study and careful coordination of the data set in use. These include:

- **Run-by-run analysis of the total number of runs** used in both the jet and photon trigger analyses. Each efficiency distribution lists how many runs contribute to the combined analysis, but in both cases, some runs are missing from the total number of active trigger runs shown in Table 1. These missing runs likely stem from processing issues and should be verified to recover the additional statistics.
- **Crossing angle consistency checks.** Although an offline vertex cut of 30 cm is appliedensuring a consistent vertex range for runs recorded at either 0 mrad or 1.5 mrad crossing anglesit is important to examine if separating runs based on these crossing angles significantly affects the resulting efficiency distributions.
- **Development of a dedicated prescale handler** for use across the collaboration. Such a tool would allow other analyses to incorporate the same scaled bits used here, applying run-by-run scaling factors in a standardized, straightforward manner.

$$E_T^{\text{iso}} = (\Sigma_{\text{EMCal, HCal}} E_T^{\Delta R<0.03}) - E_{T,\text{cluster}} \tag{2}$$

References

[1] Martin Purschke. sPHENIX Director's Review GL1/Timing System. Technical report, Brookhaven National Laboratory, 2019. URL: https://indico.bnl.gov/event/5961/contributions/27617/attachments/22412/31471/PD2_DAQ_GL1_Timingsystem_final.pdf. 1

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